

```
#include <stdio.h>
```

```
void BubbleSort(int a[], int n)
```

```
{
    int i, j, temp;

    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - 1; j++)
        {
            if(a[j] > a[j + 1])
            {
                temp = a[j];
                a[j] = a[j+1];
                a[j + 1] = temp;
            }
        }
    }
}
```

```
int main()
```

```
{
    int a[10], n, i;

    printf("Enter no. of elements: ");
    scanf("%d", &n);

    printf("Enter Elements: ");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    BubbleSort(a,n);

    printf("Sorted Array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

```
#include <stdio.h>
```

```
void SelectionSort(int a[], int n)
```

```

{
    int i, j, min, temp;

    for(i = 0; i < n - 1; i++)
    {
        min = i;
        for(j = i + 1; j < n; j++)
        {
            if(a[j] < a[min])
            {
                min = j;
            }
        }

        temp = a[i];
        a[i] = a[min];
        a[min] = temp;

        printf("Pass %d: ", i + 1);

        for(j = 0; j < n; j++)
            printf("%d ", a[j]);

        printf("\n");
    }
}

```

```

int main()
{
    int a[10], n, i;

    printf("Enter no. of elements: ");
    scanf("%d", &n);

    printf("Enter Elements: ");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    printf("\n");

    SelectionSort(a, n);

    printf("\n");
    printf("Sorted Array: ");
    for(i = 0; i < n; i++)

```

```

        printf("%d ", a[i]);

    return 0;
}

#include <stdio.h>

void InsertionSort(int a[], int n)
{
    int i, j, key;

    for(i = 1; i < n; i++)
    {
        key = a[i];
        j = i - 1;

        while(j >= 0 && a[j] > key)
        {
            a[j + 1] = a[j];
            j--;
        }

        a[j + 1] = key;

        printf("Pass %d: ", i);

        for(j = 0; j < n; j++)
        {
            printf("%d ", a[j]);
        }

        printf("\n");
    }
}

int main()
{
    int a[10], n, i;

    printf("Enter no. of elements: ");
    scanf("%d", &n);
    printf("Enter elements: ");

```

```
for(i = 0; i < n; i++)
{
    scanf("%d", &a[i]);
}
InsertionSort(a, n);
printf("Sorted Array: ");

for(i = 0; i < n; i++)
{
    printf("%d ", a[i]);
}

return 0;
}
```